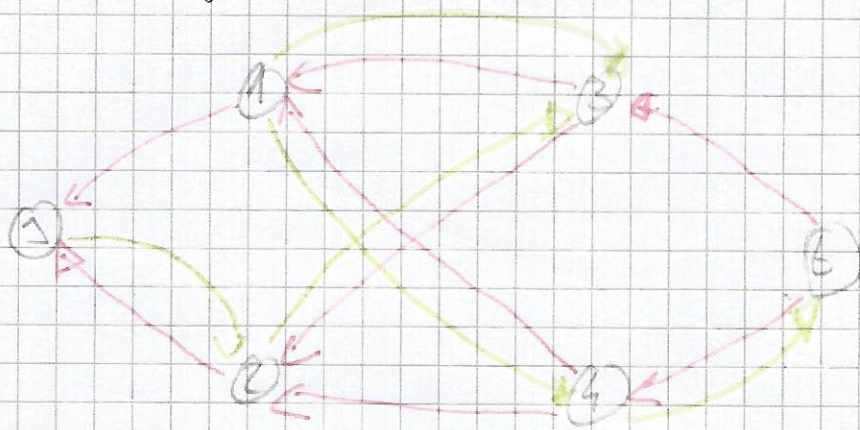


Residual graph

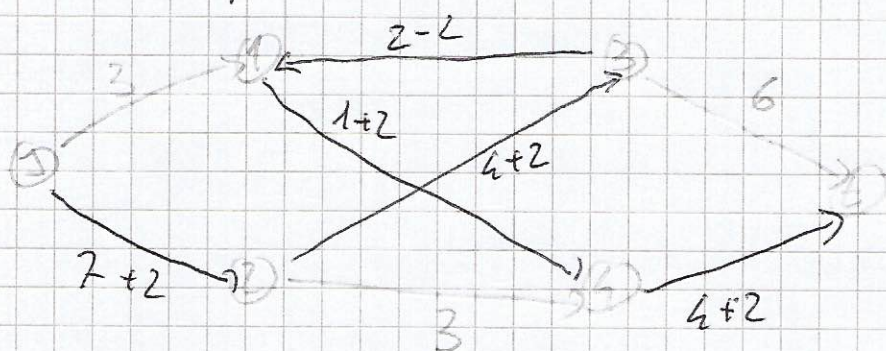


path

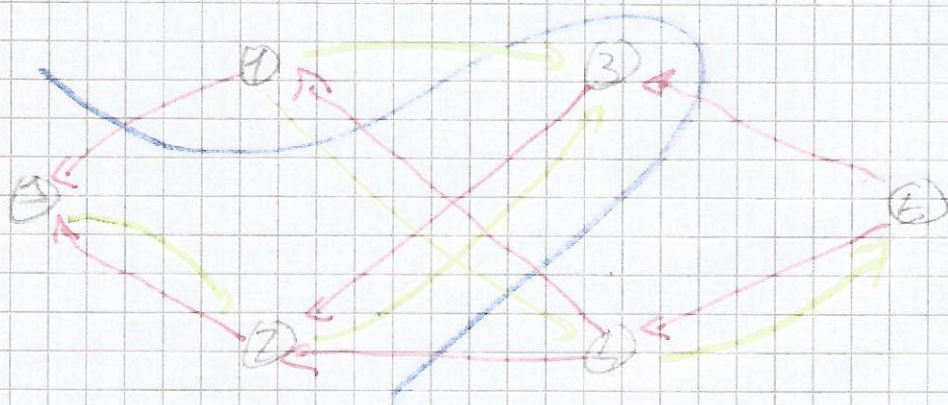
$$s \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 4 \rightarrow t$$

$$D = \min \{ 10-7, 9-4, 2, 4-1, 11-4 \} = 2$$

updated flow



New residual graph



cut $N_S = \{ s, 2, 3 \}$ $N_T = \{ 1, 4, t \}$

$$\begin{aligned}
\max \quad & 3y_1 + 3y_2 \\
& -y_1 - y_2 \leq -5 \\
& y_1 \leq 7 \\
& -2y_1 + 3y_2 \leq 10 \\
& 2y_1 - y_2 \leq 10 \\
& y_1, y_2 \geq 0
\end{aligned}$$

Complementary slackness equations

$$y_1(-x_1 + x_2 - 2x_3 + 2x_4 - 3) = 0 \quad (1)$$

$$y_2(-x_1 + 3x_3 - x_4 - 3) = 0 \quad (2)$$

$$x_1(-y_1 - y_2 + 5) = 0 \quad (3)$$

$$x_2(y_1 - 7) = 0 \quad (4)$$

$$x_3(-2y_1 + 3y_2 - 10) = 0 \quad (5)$$

$$x_4(2y_1 - y_2 - 10) = 0 \quad (6)$$

Solution $\bar{x}^1 = [0, 0, 3/4, 15/4]$

$$y_1(3 - 3) = 0 \quad y_1 = ?$$

$$y_2(3 - 3) = 0 \quad y_2 = ?$$

$$x_3^1 \neq 0$$

$$-2y_1 + 3y_2 = 10$$

$$x_4^1 \neq 0$$

$$2y_1 - y_2 = 10$$

$$\left. \begin{array}{l} -2y_1 + 3y_2 = 10 \\ 2y_1 - y_2 = 10 \end{array} \right\} \bar{y}_1^1 = 10 \quad \bar{y}_2^1 = 10$$

The complementary solution is

$$y^1 = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$$

x^1 is feasible : it satisfies the inequalities (as =) and the sign constraints

Let us check the constraints of the dual

$$-y_1 - y_2 = -20 \leq -5 \quad \text{OK}$$

$$y_1 = 10 \not\leq 7$$

VIOLATED

y^1 is not dual feasible

Solution $x'' = [0 \ 5 \ 1 \ 0]$

$$y_1(3-3) = 0 \Rightarrow y_1'' = ?$$

$$y_2(3-3) = 0 \Rightarrow y_2'' = ?$$

$$x_2' \neq 0 \Rightarrow (y_1 - 7) = 0 \quad y_1'' = 7$$

$$x_3' \neq 0 \Rightarrow (-2y_1 + 3y_2 - 10) = 0 \quad y_2'' = 8$$

the complementary solution is

$$y'' = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

x'' is feasible : it satisfies the constraints (as =) and the sign constraints

Let us check the constraints of the dual:

$$-y_1 - y_2 = -15 \leq -5 \quad \text{OK}$$

$$2y_1 - y_2 = 14 - 8 \leq 6 \leq 10 \quad \text{OK}$$

y'' satisfies also the sign const

$$\max x_1 - 5x_2$$

$$4x_1 - 12x_2 + x_3 = 15$$

$$4x_1 + 12x_2 + x_4 = 23$$

$$x_1, x_2, x_3, x_4 \geq 0$$

$$B = \{1, 4\}$$

$$A_B = \begin{bmatrix} 4 & 0 \\ 4 & 1 \end{bmatrix}$$

$$A_B^{-1} = \begin{bmatrix} 1/4 & 0 \\ -1 & 1 \end{bmatrix}$$

$$A_N = \begin{bmatrix} -12 & 1 \\ 12 & 0 \end{bmatrix}$$

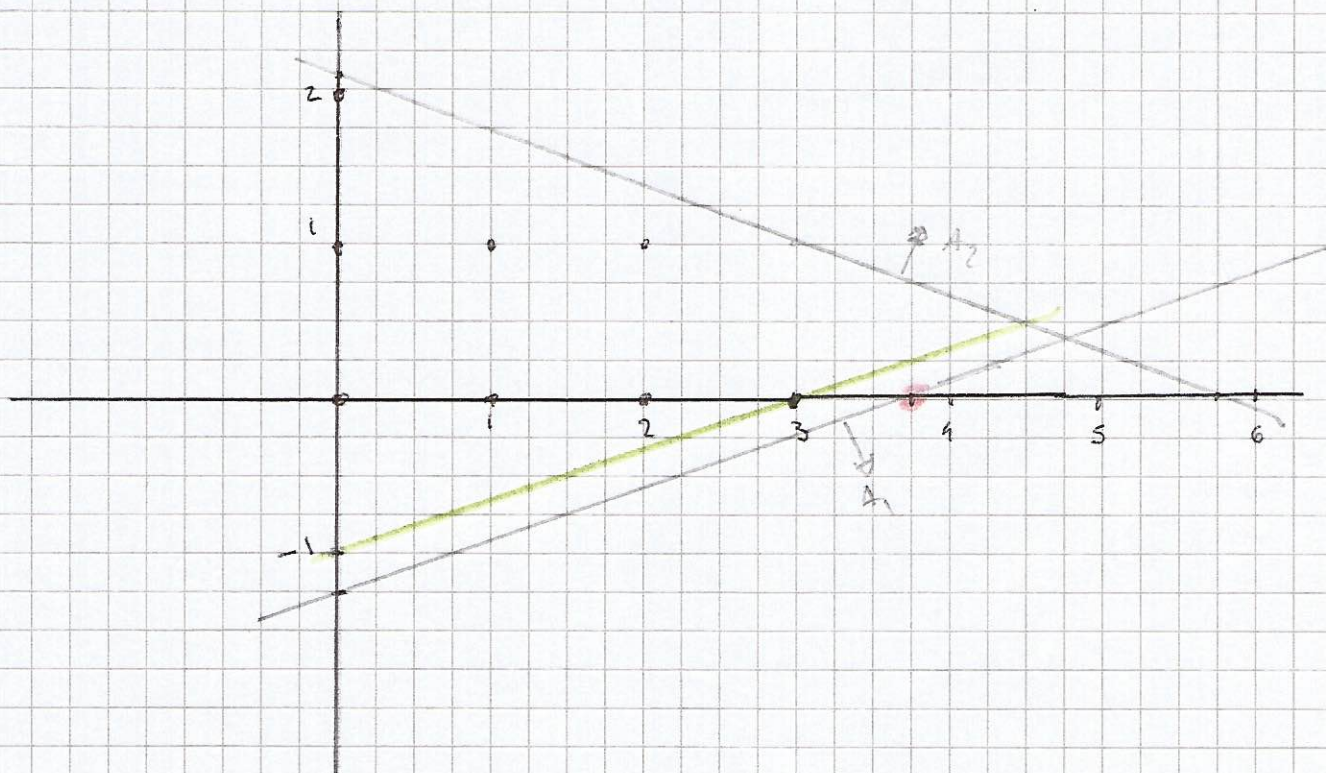
$$A_B^{-1} A_N = \begin{bmatrix} 15/4 \\ 8 \end{bmatrix}$$

$$A_B^{-1} A_N = \begin{bmatrix} -3 & 1/4 \\ 22 & -1 \end{bmatrix} \begin{matrix} x_2 & x_3 \\ x_1 & x_4 \end{matrix}$$

$$\text{cut 1} \quad x_1 - 3x_2 + \left[1/4\right]x_3 \leq \left[\frac{15}{4}\right]$$

$$x_1 - 3x_2 \leq 3$$

the other component is integer



$\{1 \dots q\}$ versions of the fake news

$\{1 \dots u\}$ profiles

N set of real users
 $F(h)$ fake profiles ~~can~~ visible by h
variables

$$y_h = \begin{cases} 1 & h \in N \text{ is affected} \\ 0 & \end{cases}$$

$$x_{ij} = \begin{cases} 1 & \text{version } i \text{ is posted by } j \\ 0 & \end{cases}$$

objective function

$$\max \sum_{h \in N} y_h$$

$$\sum_{j=1}^q x_{ij} = 1 \quad i = 1 \dots u$$

$$\sum_{j=1}^q \sum_{i \in F(h)} x_{ij} \geq K y_h \quad \forall h \in N \quad (*)$$

variant

$$z_{hj} = \begin{cases} 1 & \text{if } h \text{ sees version } j \\ 0 & \end{cases}$$

$$\sum_{j=1}^q z_{hj} \geq K y_h \quad h \in N \quad \text{instead of } (*)$$

$$z_{hj} \leq \sum_{i \in F(h)} x_{ij} \quad h \in N \quad j = 1 \dots q$$

Graph $G = (N, A)$

$$N = \{0 \dots n\}$$

representing the days of the planning horizon

$$A = \{(i, j) \mid i < j \in N\} \rightarrow \text{acyclic graph}$$

arc (i, j) represents the production from day $i+1$ to day $j-1$ and the reconditioning at day j

$$\begin{aligned} r_{ij} &= \text{revenue for selling the energy produced} \\ &\quad \text{from } i+1 \text{ to } j-1 \\ &= \sum_{l=i+1}^{j-1} c_{i+l} (b - a(l-i)) \end{aligned}$$